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!pip install -q transformers datasets evaluate sentencepiece sacrebleu rouge-score
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import torch
import pandas as pd
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM
from datasets import Dataset
from evaluate import load
from sklearn.model_selection import train_test_split
from google.colab import files
from torch.utils.data import DataLoader

device = "cuda" if torch.cuda.is_available() else "cpu"
print("Using device:", device)

uploaded = files.upload()
csv_file = list(uploaded.keys())[0]
print("Uploaded file:", csv_file)

df = pd.read_csv(csv_file)
df.columns = df.columns.str.lower()
print(df.head())
print(df.columns)

train_q, test_q, train_a, test_a = train_test_split(
    df["question"].tolist(),
    df["answer"].tolist(),
    test_size=0.3,
    random_state=42
)

train_dataset = Dataset.from_dict({
    "question": train_q,
    "answer": train_a
})

test_dataset = Dataset.from_dict({
    "question": test_q,
    "answer": test_a
})

model_name = "google/mt5-small"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name).to(device)

max_len = 128

def preprocess(batch):
    batch["question"] = ["answer: " + q for q in batch["question"]]
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inputs = tokenizer(
    batch["question"],
    padding="max_length",
    truncation=True,
    max_length=max_len
)

targets = tokenizer(
    batch["answer"],
    padding="max_length",
    truncation=True,
    max_length=max_len
)

labels = [
    [(t if t != tokenizer.pad_token_id else -100) for t in seq]
    for seq in targets["input_ids"]
]

inputs["labels"] = labels
return inputs

train_tokenized = train_dataset.map(preprocess, batched=True)
test_tokenized = test_dataset.map(preprocess, batched=True)

train_tokenized = train_tokenized.remove_columns(["question",
"answer"])
test_tokenized = test_tokenized.remove_columns(["question", "answer"])

train_tokenized.set_format("torch")
test_tokenized.set_format("torch")

train_loader = DataLoader(train_tokenized, batch_size=8, shuffle=True)
test_loader = DataLoader(test_tokenized, batch_size=1)

optimizer = torch.optim.AdamW(model.parameters(), lr=3e-5)

model.train()

for epoch in range(3):
    total_loss = 0
    for batch in train_loader:
        batch = {k: v.to(device) for k, v in batch.items()}
        optimizer.zero_grad()
        outputs = model(**batch)
        loss = outputs.loss
        loss.backward()
        optimizer.step()
        total_loss += loss.item()
    print(f"Epoch {epoch+1}/3 | Avg Loss:

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{total_loss/len(train_loader):.4f}")

model.eval()
rouge = load("rouge")
bleu = load("bleu")

predictions = []
references = []

with torch.no_grad():
    for batch in test_loader:
        input_ids = batch["input_ids"].to(device)
        outputs = model.generate(
            input_ids,
            max_new_tokens=64,
            num_beams=4,
            do_sample=False
        )
        pred = tokenizer.decode(outputs[0], skip_special_tokens=True)
        ref = tokenizer.decode(
            [x for x in batch["labels"][0].tolist() if x != -100],
            skip_special_tokens=True
        )
        predictions.append(pred)
        references.append(ref)

rouge_results = rouge.compute(
    predictions=predictions,
    references=references
)

bleu_results = bleu.compute(
    predictions=predictions,
    references=[[r] for r in references]
)

print(f"ROUGE-L Score: {rouge_results['rougeL']:.4f}")
print(f"BLEU-4 Score: {bleu_results['bleu']:.4f}")

for i in range(min(3, len(predictions))):
    print("Q :", test_q[i])
    print("Pred:", predictions[i])
    print("Ref :", test_a[i])
    print("-" * 60)

```

Using device: cuda

<IPython.core.display.HTML object>

Saving BankFAQs.csv to BankFAQs (4).csv

Uploaded file: BankFAQs (4).csv

```

                                question \
0 What are the documents required for opening a ...
1 Can I transfer my Current Account from one bra...
2 My present status is NRI. What extra documents...
3 What are the documents required for opening a ...
4 What documents are required to change the addr...

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                                answer      class
0 Following documents are required to open a Cur... accounts
1 Yes, Current Accounts can be transferred from ... accounts
2 NRI/PIO can open the proprietorship/partnershi... accounts
3 Following documents are required for opening a... accounts
4 Following documents are required to change the... accounts
Index(['question', 'answer', 'class'], dtype='object')

```

```

{"model_id": "30d9818625e54bfaalc58577f958ba13", "version_major": 2, "version_minor": 0}

```

The tied weights mapping and config for this model specifies to tie shared.weight to encoder.embed_tokens.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie\_word\_embeddings=False` to silence this warning

The tied weights mapping and config for this model specifies to tie shared.weight to decoder.embed_tokens.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie\_word\_embeddings=False` to silence this warning

The tied weights mapping and config for this model specifies to tie shared.weight to lm_head.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie\_word\_embeddings=False` to silence this warning

```

{"model_id": "2a9db4ebad324c699f93c52de48781d7", "version_major": 2, "version_minor": 0}

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```

{"model_id": "0535cb5c1e124dda91234689b18d84f0", "version_major": 2, "version_minor": 0}

```

Epoch 1/3 | Avg Loss: 18.4527

Epoch 2/3 | Avg Loss: 12.8214

Epoch 3/3 | Avg Loss: 10.6066

ROUGE-L Score: 0.0392

BLEU-4 Score: 0.0006

Q : Can I sell/redeem schemes bought offline through online

Pred: <extra_id_0>.

Ref : No, only those schemes bought through the online mode can be sold /redeemed through this service.

Q : In case of existing investors, when and how will the KYC norms be introduced

Pred: <extra_id_0>

Ref : KYC norms are applicable to all investors. It is in the interest of all investors to obtain KYC Acknowledgement and submit it to the Mutual Fund to avoid any inconvenience in the future.

Q : What is an injury or injuries

Pred: <extra_id_0>

Ref : Injury or injuries means any physical, external or accidental bodily injury occurring suddenly in time and resulting solely and independently of any other cause or any physical defect or infirmity existing before the period of insurance.
